

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer at an early-stage healthtech startup, building full-stack product features across the stack (Next.js + Firebase/GCP) for a **SMART on FHIR** clinical decision support app that surfaces covered, guideline-backed medication alternatives inside the EHR prescribing workflow to prevent prior authorization.
- Extending a **Firestore Cloud Functions** backend that builds alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with an **AI-powered workflow** where **Gemini** is constrained to synthesis while the backend deterministically controls eligibility and coverage.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR text-analysis platform (Python, **FastAPI**, sentence-transformers, React) as a self-serve full-stack web tool, productionizing a published **NLP** method and owning architecture and technical decisions end to end.
- Built versioned research-data infrastructure: a YAML model registry, a construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system over ingested corpora; grew the backend suite to **40 hermetic tests** in CI.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** through **REST** integrations with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Architected a full-stack data annotation platform (React, Node.js, MongoDB) used daily by the AI team, replacing spreadsheet workflows with a purpose-built annotation UI that cut manual annotation by **100+ hours/month** and lifted labeling throughput by **60%**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage AI system that extracts structured data from unstructured sources, turning open-ended queries into entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Scaled a data-ingestion pipeline over a **~1.4B-row** dataset, cutting end-to-end ETL runtime **~25%** (17–19 min to 14.3 min) via broadcast joins, Adaptive Query Execution, skew-join handling, shuffle-partition tuning, schema-aware reads, and year/month partitioning.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a full-stack cross-platform desktop app combining **computer vision** and **speech** on-device (MediaPipe hand tracking, gesture classification, whisper.cpp ASR, local TTS) on a FastAPI + Electron/React stack with async queues, running fully offline.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, SQL, C++, Java, MongoDB, PostgreSQL, SQLite

Frameworks & Web: React, Next.js, Node.js, Express, FastAPI, REST APIs, Docker, Git, GitHub Actions

ML & AI: PyTorch, Hugging Face Transformers, sentence-transformers, NLP, LLM agents, RAG, evaluation

Data & Retrieval: Spark / PySpark, ETL, data pipelines, cross-encoder reranking, RRF, cell-level provenance

Cloud & DevOps: AWS, GCP, Azure, Firebase, CI/CD, Linux

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, ProjectsHut. - Tutored **30+** students at Sayur, a non-profit in Tamil Nadu.